Revisiting growth dynamics in G7: an econometric critique of AI, education and industrialization interactions

Journal of Economic and Administrative Sciences

Ihsen Abid
Department of Finance, College of Business,
Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia

Received 19 May 2025
Revised 18 July 2025
Accepted 29 July 2025

Abstract

Purpose – This study explores the key determinants of GDP growth in G7 countries (Canada, France, Germany, Italy, Japan, the UK and the United States). The analysis spans annual data from 2010 to 2023, emphasizing the roles of AI innovation, education, industrialization, governance and trade in shaping economic growth within G7 economies.

Design/methodology/approach – A dynamic panel data regression model is applied using the Arellano and Bond (1991) generalized method of moments (GMM) estimator to address endogeneity, autocorrelation and heteroskedasticity. Robustness checks include re-estimations with robust standard errors and Driscoll–Kraay standard errors to correct for cross-sectional dependence and further strengthen the validity of the results.

Findings – The results reveal that industrial value added, trade openness and investment growth significantly and positively influence GDP growth in G7 economies. Conversely, AI patent applications and government expenditure on education have negative effects, which may reflect short-term inefficiencies, resource diversion or misalignment between spending and labor market demands.

Originality/value – This study contributes to the literature by providing updated empirical evidence on growth drivers in advanced economies. It highlights the importance of balancing innovation and education spending with effective policy frameworks to maximize their long-term contribution to economic growth.

Keywords G7 countries, GDP growth, AI patents, Education expenditure, Industrialization, Trade openness, Investment, Dynamic panel regression, Arellano–Bond estimation

Paper type Research article

1. Introduction

The study of economic growth and its key determinants has long been a central focus in economic research. Understanding the factors that drive GDP growth is essential for policymakers seeking to enhance economic performance and ensure sustainable development. Traditional growth theories emphasize the roles of capital accumulation, labor force expansion, and technological progress (Solow, 1956; Romer, 1990). Meanwhile, contemporary analyses incorporate institutional quality, trade openness, and policy stability as critical contributors to long-term growth (Acemoglu *et al.*, 2005; Rodrik, 2008). Despite extensive research, empirical results remain mixed due to differences in national contexts and external conditions (Barro, 1996; Easterly and Levine, 2001).

This study aims to evaluate how innovation, education, industrialization, trade openness, and investment influence GDP growth in advanced economies, particularly the G7 countries. It addresses the central research problem of whether these traditionally accepted drivers of growth continue to yield expected positive effects in mature, structurally complex economies.

JEL Classification — C33 (Panel Data Models), O40 (Economic Growth), O30 (Innovation), F43 (Economic Growth in Open Economies), H52 (Government Expenditures on Education)

Funding statement: This work was supported and funded by the Deanship of Scientific Research at Imam Mohammad Ibn Saud Islamic University (IMSIU) (grant number IMSIU-DDRSP2504).

Declaration of conflicting interest: The authors declare no conflict of interest in relation to this research.

Journal of Economic and Administrative Sciences

e-ISSN: 2054-6246
p-ISSN: 2054-6238
DOI 10.1108/JEAS-05-2025-0295

While these determinants are widely recognized as influential factors in economic performance (Lucas, 1988; Grossman and Helpman, 1991), their actual impact may diverge in developed contexts due to institutional rigidity, aging populations, or technological saturation (Aghion and Howitt, 2009).

Focusing on G7 economies, Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States, this study analyzes how these advanced nations leverage innovation, education, and trade to support growth. These countries are leaders in technology and global trade (OECD, 2019; World Bank, 2021) but also face structural challenges like demographic shifts and market volatility (Gordon, 2016; IMF, 2023).

The purpose of this research is to test whether key growth determinants retain their explanatory power in developed economies, using a robust panel data analysis. The findings will contribute to the debate on the effectiveness of conventional growth policies in high-income countries. By examining their experiences, the research offers insights into effective policy practices and potential pitfalls in growth strategies. The results aim to inform both advanced and developing economies on how to foster inclusive and resilient growth.

The rest of the paper is structured as follows: Section 2 contains a review of the relevant literature. Section 3 explains the data sources and methodology used in the analysis. Section 4 presents the empirical results, including diagnostic tests and regression findings. Section 5 discusses the implications of the results in the context of existing literature and economic theory. Section 6 concludes the paper and offers policy recommendations based on the findings.

2. Literature review

The relationship between GDP growth and its determinants has been extensively studied in the economic literature, with a focus on factors such as innovation, education, industrialization, trade openness, investment, institutional quality, and policy stability. However, the findings are often mixed, reflecting the complexity of economic growth dynamics and the varying contexts of different studies. Although numerous studies have explored these determinants, the literature still lacks a comprehensive, context-sensitive understanding of how these variables interact under different institutional and policy environments, particularly across high-income and developing regions. Moreover, many recent developments, including the role of healthcare expenditure, external debt dynamics, and governance quality, remain underexplored in traditional growth frameworks.

This literature review explores existing research on these key determinants, highlighting studies that both validate and challenge the potential relationships examined in this paper.

2.1 Innovation and GDP growth

Innovation, particularly in the form of patent applications and technological advancements, is widely regarded as a driver of economic growth. Schumpeter (1942) introduced the concept of creative destruction, emphasizing that innovation fuels economic progress by replacing outdated technologies. Romer (1990) and Aghion and Howitt (1992) argue that technological progress, driven by innovation, is a key engine of long-term growth. However, the short-term effects of innovation can be disruptive. Acemoglu and Restrepo (2020) find that automation and AI technologies may displace labor in the short run, leading to economic disruptions before productivity gains are realized. Gordon (2016) suggests that the economic impact of recent technological advancements may be weaker than past industrial revolutions, questioning the sustainability of innovation-led growth. Similarly, Brynjolfsson *et al.* (2019) highlight that the benefits of AI innovation often take years to materialize, as firms and workers adapt to new technologies. Studies on G7 countries, such as Griffith *et al.* (2004), show that R&D investment and innovation are crucial for sustaining growth in advanced economies, but the returns to innovation depend on absorptive capacity and policy frameworks. In this context, Abid (2025a) provides compelling evidence from East Asia that artificial intelligence (AI) innovation plays a significant role in boosting economic performance, particularly when combined with institutional readiness and technological infrastructure. Similarly, Abid *et al.* (2024a) argue that macroeconomic policy effectiveness, often shaped by political stability, influences innovation outcomes in MENA economies.

These studies suggest that while innovation is crucial for long-term growth, its immediate impact on GDP growth may be negative due to adjustment costs and resource diversion.

2.2 Education and GDP growth

Education is often considered a cornerstone of human capital development and economic growth. Barro (1991) and Mankiw *et al.* (1992) emphasize the positive relationship between education and GDP growth, arguing that educated populations are more productive and innovative. However, the quality of education spending matters more than the quantity. Hanushek and Woessmann (2020) show that poorly managed education systems can lead to diminishing returns, where increased spending does not translate into improved human capital or economic growth. Krueger and Lindahl (2001) suggest that the relationship between education and growth is nonlinear, with diminishing returns at higher levels of education. Empirical studies on G7 countries (OECD, 2021) highlight that investment in higher education and vocational training significantly contributes to labor productivity and long-term growth, with Germany and Canada showing particularly strong links between education and economic performance. Additionally, Becker (1964) highlights the role of on-the-job training as a crucial complement to formal education in driving productivity. Abid (2025b) contributes to this literature by examining labor force participation trends in Gulf Cooperation Council (GCC) countries, identifying education and skill development as key enablers of higher economic participation and growth.

These studies suggest that the relationship between education expenditure and GDP growth is not always straightforward and may depend on the effectiveness of education policies.

2.3 Industrialization and GDP growth

Industrialization has long been recognized as a key driver of economic growth. Rodrik (2004) highlights the unique spillover effects of manufacturing sectors, such as technology transfer and skill development, which contribute to sustained economic growth. Chang (2002) argues that industrial policy, including state-led investment in key industries, has played a critical role in the rapid growth of East Asian economies. Szirmai and Verspagen (2015) find a strong positive correlation between industrial growth and GDP growth, particularly in advanced economies. However, Dasgupta and Singh (2006) caution that the benefits of industrialization may be unevenly distributed, with some sectors or regions benefiting more than others. In the context of G7 countries, Fagerberg and Verspagen (2002) demonstrate that industrial productivity growth has been a major determinant of overall economic expansion, particularly in manufacturing-intensive economies like Japan and Germany. Moreover, de-industrialization in some advanced economies has raised concerns about whether services can fully replace manufacturing as a growth engine (Rodrik, 2015). Abid (2025c) analyzes the balance between economic growth and sustainability in South Asia and finds that industrial policy, if aligned with environmental constraints, can still serve as a driver of growth through green industrialization. Likewise, Abid (2025d), focusing on South Africa, illustrates how technological shifts in industry affect employment levels, which has downstream implications for GDP growth.

These studies suggest that while industrialization is generally associated with positive economic outcomes, its impact may vary depending on structural and institutional factors.

2.4 Trade openness and GDP growth

Trade openness is widely regarded as a catalyst for economic growth. Frankel and Romer (2019) demonstrate that trade significantly boosts GDP growth by increasing efficiency, specialization, and access to larger markets. Dollar and Kraay (2004) provide empirical evidence that trade liberalization contributes to poverty reduction and economic growth, particularly in developing economies. The World Bank (2021) reports that countries with higher trade openness tend to experience faster economic growth due to increased foreign investment and innovation. However, Rodrik (2018) argues that the benefits of trade openness may be offset by negative effects, such as increased inequality and vulnerability to external shocks. Studies on G7 economies (Baldwin and Taglioni, 2006) indicate that trade integration within the EU and North America has significantly enhanced economic performance, but rising protectionism in recent years poses risks to sustained growth. Krugman (1994) suggests that while free trade can enhance efficiency, strategic trade policies may be necessary to protect nascent industries in developing economies. Abid (2025e) explores the role of international tourism as a form of trade in services and finds that institutional quality and technological infrastructure significantly affect trade-related growth in MENA countries.

These studies suggest that while trade openness is generally beneficial for GDP growth, its impact may depend on complementary policies and institutional frameworks.

2.5 Investment and GDP growth

Investment in physical capital, such as infrastructure and machinery, is a critical driver of economic growth. Barro and Sala-i-Martin (2017) emphasize that capital accumulation enhances productivity and output, particularly in advanced economies. Lucas (1988) introduces the concept of endogenous growth, arguing that investment in human and physical capital creates positive externalities that sustain long-term growth. The IMF (2022) highlights the importance of investment in infrastructure and technology for sustaining long-term economic growth. However, Easterly and Levine (2001) caution that investment alone is not sufficient for growth and must be accompanied by sound institutions and policies. For G7 countries, Blanchard *et al.* (2002) find that private sector investment, especially in technology-intensive industries, has been a significant contributor to GDP growth, with varying effects depending on national regulatory frameworks. Moreover, Levine and Zervos (1998) find that financial development plays a crucial role in ensuring that investment translates into economic growth. Abid and Gafsi (2025) emphasize the interconnected role of economic complexity, technological integration, and environmental sustainability in Saudi Arabia, underlining the importance of long-term investment strategies to enhance GDP growth.

These studies suggest that while investment is a key determinant of GDP growth, its effectiveness depends on the broader economic and institutional context.

2.6 Institutional and fiscal determinants

Recent research has emphasized new dimensions in the growth literature, particularly in developing regions. For instance, Abdulqadir *et al.* (2022) find that healthcare expenditure plays a dynamic and positive role in supporting GDP growth in Sub-Saharan Africa, suggesting that social sector investments can yield macroeconomic returns. Similarly, Muthoni and Osoro (2023) demonstrate that while external debt can stimulate growth through public investment, unsustainable debt burdens can offset these benefits. Abid (2025f) shows that employment dynamics and macroeconomic variables such as inflation, interest rates, and investment play a significant role in shaping long-term economic trends in MENA countries, reinforcing the need for region-specific fiscal strategies. Governance quality also matters: Beyene (2022) shows that strong governance institutions significantly moderate the effectiveness of macroeconomic policy and amplify economic growth. Abid *et al.* (2024b) find that energy intensity and CO_2 emissions significantly impact GDP growth in Gulf countries, illustrating the need for sustainable institutional planning.

Abid (2025g) also confirms that environmental and economic variables jointly influence green growth in Saudi Arabia, underscoring the role of policy frameworks. Furthermore, Chaabouni and Abid (2025) explore how institutional energy consumption patterns influence macroeconomic performance in the GCC region, reinforcing the significance of policy context.

These studies point to the necessity of including institutional and social infrastructure factors in any holistic assessment of economic growth determinants.

In conclusion, the literature on GDP growth and its determinants reveals a complex interplay of factors, with mixed findings across studies. While innovation, education, industrialization, trade openness, and investment are generally regarded as positive drivers of growth, their impacts can vary depending on contextual factors such as institutional quality, policy effectiveness, and time horizons. The inclusion of recent studies on healthcare, governance, and external debt further refines the understanding of the multidimensional drivers of growth, especially in developing contexts. This literature review underscores the need for a nuanced understanding of the drivers of economic growth and the importance of context-specific policies to maximize their potential benefits.

3. Methodology

3.1 Data

The dataset includes economic and innovation-related variables for G7 countries (Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States) from 2010 to 2023. To ensure the accuracy and validity of the analysis, the study data are retrieved from the World Bank and are transformed in logarithm (excluding the variables in growth form). Table 1 presents a description of the variables and their classification.

The dependent variable, GDP growth (annual %), measures the annual percentage growth rate of GDP and serves as the primary indicator of economic performance.

The independent variables include patent applications per 1 million people related to artificial intelligence (AI), which reflects technological innovation; government expenditure on education as a percentage of GDP, indicating investment in human capital; industry value added as a percentage of GDP, representing the contribution of the industrial sector; trade as a percentage of GDP, measuring economic openness; and gross capital formation growth, reflecting investment in fixed assets. These variables collectively provide insights into the drivers of economic growth, such as innovation, education, industrialization, trade, and investment in G7 countries.

3.2 Econometric modeling

- *Panel unit root tests*

To ensure the stationarity of the variables and prevent spurious regression results, this study employs the Cross-Sectionally Augmented IPS (CIPS) test proposed by Pesaran (2007). The

Table 1. Data description

Classification	Variable	Description
Economic growth indicator	GDPG	GDP growth (annual %)
Innovation and technological development indicator	PAT	Patent applications - Artificial-intelligence-patents-submitted-per-million
Public investment and human capital indicator	GEE	Government expenditure on education, total (% of GDP)
Sectoral contribution to GDP indicator	IND	Industry (including construction), value added (% of GDP)
Trade and economic openness indicator	TRA	Trade (% of GDP)
Investment and capital formation indicator	GCFG	Gross capital formation (annual % growth)

Source(s): Author’s own work

CIPS test accounts for cross-sectional dependence by incorporating the cross-sectional averages of the dependent variable into the unit root testing procedure.

- *Cross-sectional dependence tests*

Given the potential for economic interdependence among G7 countries, cross-sectional dependence (CSD) is assessed using three established tests. First, Pesaran's (2007) CD test provides a robust approach to detecting cross-sectional dependence in panel data settings. Second, Friedman's (1937) non-parametric test evaluates dependence without assuming normality. Third, Frees' (1995) rank-based test assesses cross-sectional dependence using the correlation of residuals across panel units. These complementary tests help ensure the reliability of subsequent panel estimations by accounting for potential inter-country linkages.

- *Slope heterogeneity test*

To assess the homogeneity of slope coefficients across G7 countries, the Pesaran and Yamagata (2008) slope heterogeneity test is applied. This test evaluates whether the slope coefficients are identical across cross-sections, which is crucial for selecting an appropriate panel data estimator.

•. *Panel cointegration tests*

To investigate the long-run relationship between GDP growth and the independent variables, panel cointegration is tested using three complementary methods. The Kao (1999) test assesses the presence of a common long-run equilibrium relationship across panel units. The Pedroni (1999, 2004) test offers several statistics that account for heterogeneity in cointegration dynamics across countries. Lastly, the Westerlund (2008) test, based on error correction models, evaluates whether panel members exhibit cointegration by testing for the significance of adjustment toward equilibrium. Together, these tests enhance the robustness of long-run relationship detection in the panel setting.

- *Panel data regression model*

Given the dynamic nature of economic growth, the Arellano and Bond (1991) dynamic panel-data estimation technique is employed. This generalized method of moments (GMM) approach accounts for endogeneity, autocorrelation, and heteroskedasticity, ensuring consistent and unbiased estimates. The estimated model is specified as follows:

$$GDPG_{it} = \alpha + \beta_1 GDPG_{it-1} + \beta_2 PAT_{it} + \beta_3 GEE_{it} + \beta_4 IND_{it} + \beta_5 TRA_{it} + \beta_6 GCFG_{it} + \varepsilon_{it} \tag{1}$$

where:

$GDPG_{it}$ represents GDP growth for country at time,

$GDPG_{it-1}$ is the lagged dependent variable to control for persistence in economic growth,

PAT_{it} denotes patent applications per million people related to AI,

GEE_{it} represents government expenditure on education as a percentage of GDP,

IND_{it} indicates industry value added as a percentage of GDP,

TRA_{it} measures trade as a percentage of GDP,

$GCFG_{it}$ signifies gross capital formation growth,

ε_{it} is the error term.

• *Robustness checks*

To ensure the robustness of the empirical findings, diagnostic tests are applied to assess potential econometric issues. The Wooldridge (2002) test is employed to detect first-order autocorrelation in panel data, which can bias standard errors and inference. Additionally, a heteroskedasticity test is conducted to identify unequal variances in the error terms across observations, which may affect the efficiency of estimators. These robustness checks help confirm the reliability and validity of the estimated results.

Additionally, the model is re-estimated using the Arellano-Bond dynamic panel-data estimation with robust standard errors and Driscoll-Kraay standard errors to correct for cross-sectional dependence and heteroskedasticity.

To reinforce the robustness of our findings, we re-estimate the baseline model using an alternative long-run estimator: Fully Modified Ordinary Least Squares (FMOLS). This method is well-suited for panel data characterized by potential endogeneity and serial correlation, offering more reliable coefficient estimates. FMOLS is widely recognized in empirical growth literature for its ability to correct for these issues, as highlighted by Pedroni (1999).

4. Results

4.1 Data analysis

The correlation matrix shows the pairwise relationships between the dependent and the independent variables (Table 2).

GDP growth is strongly correlated with gross capital formation growth (0.7970), highlighting investment as a key growth driver. Other variables, including AI patent applications, industry value added, trade, and education spending, show weak or negligible correlations with GDP growth. Government expenditure on education has a weak positive correlation with GDP growth (0.0890) but a strong negative correlation with industry value added (−0.7034), suggesting a possible trade-off. Trade shows a moderate negative correlation with patent applications (−0.3568) and a moderate positive correlation with education spending (0.2400).

The descriptive statistics reveal key characteristics of the dataset (Table 3).

Descriptive statistics (Table 3) show that GDP growth averages 1.37% with high volatility (SD = 3.05), while gross capital formation growth also varies widely (mean = 1.99%, SD = 5.68). Other variables such as trade (3.97%), industry value added (3.06%), and education expenditure (1.54%) are more stable. AI patent applications vary significantly across countries (mean = 0.91, range = −4.45 to 3.72).

4.2 Panel unit root tests

Table 4 provides the results of the Cross-Sectionally Augmented IPS (CIPS) test for stationarity, which is used to determine whether the variables in the dataset have a unit root

Table 2. Correlation matrix

Variable	GDPG	PAT	GEE	IND	TRA	GCFG
GDPG	1.0000					
PAT	−0.0369	1.0000				
GEE	0.0890	−0.0106	1.0000			
IND	−0.0648	0.1720	−0.7034	1.0000		
TRA	−0.0083	−0.3568	0.2400	0.0700	1.0000	
GCFG	0.7970	−0.0208	0.0721	−0.0501	−0.0160	1.0000

Source(s): Author's own work

Table 3. Descriptive statistics

Variable	Obs	Mean	Std. dev.	Min	Max
GDPG	77	1.3696	3.0456	−10.2969	8.9311
PAT	77	0.9094	1.8857	−4.4528	3.7153
GEE	77	1.5420	0.1826	1.1242	1.8293
IND	77	3.0613	0.1914	2.7781	3.3763
TRA	77	3.9706	0.3653	3.1390	4.4894
GCFG	77	1.9889	5.6830	−11.5255	28.2390

Source(s): Author's own work

Table 4. Cross-Sectionally Augmented IPS (CIPS) test

Variable	CIPS statistic	Critical value	Variable	CIPS statistic	Critical value
GDPG	−2.3810	−2.2200	ΔGDPG	−3.1670	−2.3100
PAT	−3.6710	−2.2000	ΔPAT	−2.7750	−2.3100
GEE	−2.0870	−2.2200	ΔGEE	−3.2890	−2.2800
IND	−2.0740	−2.2000	ΔIND	−2.7760	−2.2800
TRA	−1.1870	−2.2200	ΔTRA	−3.0530	−2.3100
GCFG	−1.8940	−2.2200	ΔGCFG	−2.5290	−2.3100

Source(s): Author's own work

(non-stationary) or not (stationary). The test is applied to both the level variables and their first differences (differenced variables).

GDPG and PAT are stationary at levels, while GEE, IND, TRA, and GCFG are stationary only after first differencing. First differences (Δ) are thus used for these variables to ensure valid regression analysis.

4.3 Panel cross-sectional dependence estimations

Table 5 presents the results from the cross-sectional dependence test across the G7 countries.

All tests (Pesaran CD, Frees, Friedman) confirm strong cross-sectional dependence among G7 countries, suggesting economic shocks in one country significantly affect others due to close integration.

4.4 Test for slope heterogeneity

The Pesaran and Yamagata (2008) slope heterogeneity test is used to determine whether the slope coefficients in a panel data model are homogeneous (the same across all cross-sectional units) or heterogeneous (vary across units) (Table 6).

Test results support homogeneous slopes across countries, indicating a consistent relationship between GDP growth and its determinants in the G7.

Table 5. Cross-sectional dependence tests

Test	Statistic	*p*-value
Pesaran's test	6.6460	0.0000
Friedman's test	27.5580	0.0001
Frees' test	0.4120	10%: 0.2333, 5%: 0.3103, 1%: 0.4649

Source(s): Author's own work

Table 6. Slope heterogeneity test

Statistic	Value	*p*-value
Delta	−0.3000	0.7640
Adjusted delta	−0.4980	0.6190

Source(s): Author's own work

4.5 Panel co-integration tests

The results from the Kao (1999), Pedroni (1999, 2004), and Westerlund (2008) cointegration tests provide insights into whether a long-run relationship exists among the variables in the G7 panel data (Table 7).

The Pedroni test supports the existence of a long-run relationship, Kao results are mixed, and Westerlund shows no cointegration. Overall, evidence of cointegration is inconclusive.

4.6 Panel data regression

The results from the *Arellano-Bond dynamic panel-data estimation* provide insights into the relationship between GDP growth and the explanatory variables while accounting for dynamics and potential endogeneity (see Table 8).

Table 7. Cointegration test results

Test	Statistic	*p*-value
Kao (1999)		
Modified Dickey-Fuller t	−0.0308	0.4877
Dickey-Fuller t	−3.2802	0.0005
Augmented Dickey-Fuller t	−1.3803	0.0837
Unadjusted modified Dickey-Fuller t	−6.2803	0.0000
Unadjusted Dickey-Fuller t	−7.1463	0.0000
Pedroni (1999, 2004)		
Modified Phillips-Perron t	3.2119	0.0007
Phillips-Perron t	−4.3023	0.0000
Augmented Dickey-Fuller t	−3.7145	0.0001
Westerlund (2008)		
Variance ratio	0.4560	0.3242

Source(s): Author's own work

Table 8. Arellano-Bond dynamic panel-data estimation

Variable	Coefficient	Std. err.	*z*-value	*p*-value	95% confidence interval
L1.GDPG	−0.0513	0.1487	−0.3400	0.7300	[−0.3428, 0.2402]
PAT	−0.5137	0.1696	−3.0300	0.0020	[−0.8461, −0.1814]
ΔGEE	−11.5977	4.9685	−2.3300	0.0200	[−21.3357, −1.8596]
ΔIND	17.1378	6.5547	2.6100	0.0090	[4.2908, 29.9847]
ΔTRA	15.5914	3.6694	4.2500	0.0000	[8.3996, 22.7833]
ΔGCFG	0.2052	0.0531	3.8700	0.0000	[0.1012, 0.3092]
_cons	−50.5817	20.0032	−2.5300	0.0110	[−89.7872, −11.3763]

Source(s): Author's own work

The *Arellano-Bond dynamic panel-data estimation* results reveal several key insights into the drivers of GDP growth among G7 countries. The lagged GDP growth rate has no statistically significant effect on current GDP growth, indicating a lack of persistence in growth rates. AI patent applications show a significant negative relationship with GDP growth, suggesting that a 1-unit increase in AI patents per 1 million people is associated with a *0.51% decrease* in GDP growth. This counterintuitive result may reflect the time lag between innovation and economic impact or potential resource diversion. Government expenditure on education also has a significant negative effect, with a 1-unit increase associated with an *11.60% decrease* in GDP growth, which is unexpected and may warrant further investigation. On the other hand, industry value added, and trade openness have strong positive effects, with a 1-unit increase in each associated with a *17.14%* and *15.59% increase* in GDP growth, respectively, highlighting the importance of industrialization and trade for economic performance. Gross capital formation growth also positively impacts GDP growth, with a 1-unit increase associated with a *0.21% increase*, underscoring the role of investment in driving growth.

4.7 Robustness checks

Robustness checks and additional analysis are recommended to validate these findings. The results from the *Wooldridge test for autocorrelation* and the *heteroscedasticity test* provide insights into the presence of autocorrelation and heteroscedasticity in the panel data model (see Table 9).

The Wooldridge test ($p = 0.2233$) indicates no autocorrelation, but heteroscedasticity is present ($p = 0.0001$).

While the model is free from autocorrelation, the presence of heteroscedasticity requires corrective measures. To address this, *robust standard errors* estimation should be used to ensure valid and reliable results (see Table 10).

The *Arellano-Bond dynamic panel-data estimation* with Robust Standard Errors results reveal several key insights into the drivers of GDP growth among G7 countries. The lagged GDP growth rate has no statistically significant effect on current GDP growth, indicating a lack of persistence in growth rates. AI patent applications show a significant negative

Table 9. Robustness checks

Test	Null hypothesis (H_0)	Statistic	*p*-value
Wooldridge test for autocorrelation	No first-order autocorrelation	1.8440	0.2233
Heteroscedasticity test	Homoscedasticity (constant variance)	5.7900	0.0001

Source(s): Author's own work

Table 10. Arellano-Bond dynamic panel-data estimation with robust standard errors

Variable	Coefficient	Std. err.	*z*-value	*p*-value	95% confidence interval
L1.GDPG	−0.0513	0.1058	−0.4800	0.6280	[−0.2587, 0.1562]
PAT	−0.5137	0.1358	−3.7800	0.0000	[−0.7799, −0.2476]
ΔGEE	−11.5977	3.7857	−3.0600	0.0020	[−19.0175, −4.1778]
ΔIND	17.1378	3.1813	5.3900	0.0000	[10.9025, 23.3730]
ΔTRA	15.5914	6.1619	2.5300	0.0110	[3.5144, 27.6685]
ΔGCFG	0.2052	0.0533	3.8500	0.0000	[0.1008, 0.3096]
_cons	−50.5817	10.5616	−4.7900	0.0000	[−71.2821, −29.8813]

Source(s): Author's own work

relationship with GDP growth, suggesting that a 1-unit increase in AI patents per 1 million people is associated with a *0.51% decrease* in GDP growth. This counterintuitive result may reflect the time lag between innovation and economic impact or potential resource diversion. Government expenditure on education also has a significant negative effect, with a 1-unit increase linked to an *11.60% decrease* in GDP growth, which is unexpected and warrants further investigation. On the other hand, industry value added, and trade openness have strong positive effects, with a 1-unit increase in each associated with a *17.14* and *15.59% increase* in GDP growth, respectively, highlighting the importance of industrialization and trade for economic performance. Gross capital formation growth also positively impacts GDP growth, with a 1-unit increase linked to a *0.21% increase*, underscoring the role of investment in driving growth. Overall, the results suggest that while industrialization, trade, and investment are key drivers of GDP growth, AI patents and education spending has a negative effect on economic growth.

Since the panel exhibits *cross-sectional dependence*, Arellano-Bond is not efficient. Instead, we must use *Driscoll-Kraay standard errors*, which correct for heteroskedasticity and cross-dependence (see Table 11).

The regression with *Driscoll-Kraay standard errors* provides robust estimates of the relationship between GDP growth and the explanatory variables, accounting for cross-sectional dependence and heteroscedasticity. The model explains approximately 69.91% of the within-country variation in GDP growth and is statistically significant (p-value = 0.0047). The results show that a 1-unit increase in AI patent applications is associated with a *0.53% decrease* in GDP growth, suggesting that AI innovation may not directly contribute to growth in the short run. Government expenditure also has a significant negative effect, with a 1-unit increase linked to a *10.45% decrease* in GDP growth, which is counterintuitive and warrants further investigation. On the other hand, industry value added, and trade openness have strong positive effects, with a 1-unit increase associated with a *14.82* and *14.95% increase* in GDP growth, respectively, highlighting the importance of industrialization and trade for economic performance. Gross capital formation growth also positively impacts GDP growth, with a 1-unit increase linked to a *0.22% increase*, underscoring the role of investment in driving growth. Overall, the results suggest that policies promoting industrialization, trade, and investment are likely to boost GDP growth, while the negative effects of AI patents and education spending require further exploration to understand their underlying mechanisms.

4.8 Robustness checks using alternative estimators

To ensure the robustness of our findings, we re-estimate the baseline model using an alternative long-run estimators: Fully Modified Ordinary Least Squares (FMOLS).

The results, presented in Table 12, are broadly consistent with the GMM estimates. Industrialization and trade openness retain positive and statistically significant effects on GDP growth, while government education expenditure and AI patents show negative signs. These cross-method results provide additional confidence in the validity of our findings.

Table 11. Estimation with Driscoll-Kraay standard errors

Variable	Coefficient	Std. err.	t-value	p-value	95% confidence interval
PAT	−0.5280	0.1328	−3.9800	0.0030	[−0.8284, −0.2276]
ΔGEE	−10.4456	2.4834	−4.2100	0.0020	[−16.0634, −4.8278]
ΔIND	14.8168	3.9977	3.7100	0.0050	[5.7733, 23.8603]
ΔTRA	14.9488	2.8325	5.2800	0.0010	[8.5412, 21.3564]
ΔGCFG	0.2202	0.0384	5.7300	0.0000	[0.1333, 0.30704]
_cons	−43.5198	12.3186	−3.5300	0.0060	[−71.3864, −15.6531]

Source(s): Author’s own work

Table 12. FMOLS estimation results

Variable	Coefficient	Std. error	*t*-statistic	*p*-value	95% confidence interval
PAT	−0.5280	0.1352	−3.9000	0.0080	[−0.8589, −0.1971]
ΔGEE	−10.4456	3.9804	−2.6200	0.0390	[−20.1854, −0.7059]
ΔIND	14.8168	4.0842	3.6300	0.0110	[4.8230, 24.8106]
ΔTRA	14.9488	6.0211	2.4800	0.0480	[0.2158, 29.6818]
ΔGCFG	0.2202	0.0507	4.3400	0.0050	[0.0961, 0.3442]
_cons	−43.5198	12.4534	−3.4900	0.0130	[−73.9922, −13.0474]

Source(s): Author’s own work

5. Discussion

The results of the regression with reveal several key insights into the drivers of GDP growth in G7 countries, supported by recent empirical studies and economic theories.

The negative coefficient for AI patent applications (pat) suggests that an increase in AI patents per 1 million people is associated with a decrease in GDP growth. This counterintuitive result may reflect the time lag between innovation and measurable economic impact, a theme supported in the literature. While AI patents represent technological advancement, their immediate effect on GDP growth may be negative due to resource diversion, labor displacement, and adjustment costs (Parteka and Kordalska, 2023; Gonzales, 2023). Acemoglu and Restrepo (2020) argue that automation and AI can disrupt labor markets before gains in productivity are realized, while Brynjolfsson *et al.* (2019) emphasize that it may take years before firms and workers adapt to new technologies. This aligns with the theoretical perspective of Schumpeterian creative destruction (Schumpeter, 1942), where the short-term economic cost of innovation may obscure its long-run benefits. Moreover, Abid (2025a) confirms that AI innovation’s effectiveness is highly dependent on institutional readiness and policy context, further supporting the interpretation of delayed or uneven returns.

The negative coefficient for government education expenditure suggests that an increase in education spending is associated with a decrease in GDP growth. This finding appears surprising at first but reflects nuances widely acknowledged in the literature. Hanushek and Woessmann (2020) argue that the quality rather than the quantity of education spending determines its effectiveness. If resources are poorly allocated or systems are inefficient, increased expenditure may not yield higher productivity. Krueger and Lindahl (2001) suggest that inefficiencies in education systems and diminishing returns at higher education levels can undermine economic impact. This finding is also in line with Barro (1991) and Mankiw *et al.* (1992), who stress that education’s positive impact depends heavily on how spending translates into human capital. In the G7 context, where education systems are already well-developed, marginal increases in spending may yield lower returns. Abid (2025b) also finds that educational investment must align with labor market demands to drive economic participation, a misalignment that may partly explain the negative relationship observed here.

In contrast, the positive coefficient for industrial value added underscores that industrialization remains a critical driver of economic growth. This finding aligns strongly with Rodrik (2004) and Szirmai and Verspagen (2015), who highlight that manufacturing sectors provide unique spillover benefits, such as technology diffusion, skill development, and productivity enhancements. Fagerberg and Verspagen (2002) further confirm that industrial productivity has historically underpinned growth in manufacturing-intensive G7 economies such as Japan and Germany. The results corroborate Abid (2025c, d), who demonstrates that industrial policy, particularly when integrated with environmental sustainability and technological shifts, can yield significant GDP growth. These findings reaffirm the sustained importance of industrial sectors in advanced economies.

The positive coefficient for trade openness supports the widely held view that trade integration fosters economic growth. Frankel and Romer (2019) and World Bank (2021) show that open trade regimes increase market access, competition, and foreign investment, all of which drive GDP growth. However, Rodrik (2018) notes that benefits from trade openness may be uneven, with risks of external shocks and inequality. Nonetheless, for G7 economies, where institutional frameworks and absorptive capacity are robust, the growth-enhancing effects dominate. Abid (2025e) further validates that trade, particularly in services such as tourism, contributes significantly to GDP growth when institutional quality supports integration. The results thus align with both classical and contemporary evidence on trade liberalization as a growth enabler.

The positive coefficient for gross capital formation growth affirms the traditional economic theory that investment in physical capital stimulates economic growth. This finding is consistent with Barro and Sala-i-Martin (2017) and Lucas (1988), who emphasize capital accumulation and its externalities in productivity and output. Empirical support from Blanchard *et al.* (2002) and IMF (2022) underscores the role of private and public investment, particularly in infrastructure and technology, in sustaining growth across G7 economies. Abid and Gafsi (2025) also stress that in contexts like Saudi Arabia, long-term investment strategies that incorporate technology and environmental sustainability are essential for economic advancement. While this study focuses on G7 countries, the underlying investment-growth relationship appears robust across different contexts when supported by effective policy environments.

Overall, the findings of this study are largely in line with the broader literature but also offer critical refinements. While industrialization, trade openness, and investment show strong positive effects on GDP growth consistent with theoretical and empirical expectations, the negative impacts of AI patents and education expenditure call attention to the importance of timing, policy design, and institutional context. These results highlight that growth determinants do not operate in isolation and their effectiveness is contingent upon absorptive capacity, governance quality, and policy alignment, as emphasized by Muthoni and Osoro (2023). They also reinforce the idea that economic policy must be context-sensitive, especially in advanced economies where the marginal returns to traditional growth inputs may diminish without complementary reforms.

In summary, this study contributes to the growth literature by offering a context-specific analysis of G7 countries, confirming many established relationships while also challenging simplistic assumptions about innovation and education. Future research should continue exploring these nuanced interactions, particularly as the global economy adapts to technological disruptions and structural shifts.

6. Conclusion and recommendations

This study examined the drivers of GDP growth in G7 countries using robust econometric techniques, including regression with Driscoll-Kraay standard errors and Arellano-Bond dynamic panel-data estimation. The results highlight the significant roles of industrialization, trade openness, and investment in fostering economic growth, while revealing counterintuitive findings regarding the negative impacts of AI patents and government expenditure on education. Industrialization and trade openness emerged as strong positive drivers of GDP growth, consistent with existing literature that emphasizes their role in enhancing productivity, innovation, and market access. Investment in physical capital also showed a positive and significant impact, underscoring the importance of infrastructure and technological advancements for sustained economic growth. However, the negative effects of AI patents and education spending suggest potential inefficiencies, time lags, or misallocation of resources, which warrant further investigation. These findings underscore the complexity of economic growth dynamics and the need for targeted policies to maximize the benefits of innovation and human capital development.

A key limitation of this study lies in its deliberately parsimonious model specification, which excludes other potentially relevant variables such as energy consumption, labor market conditions, and institutional quality. This approach was adopted to preserve model interpretability, ensure robustness across countries and time, and avoid multicollinearity. Additionally, although dynamic panel estimators such as Arellano–Bond are suitable for short panels, they may be limited under strong cross-sectional dependence, as observed in this study. This constraint is acknowledged, and future research should consider system GMM or other robust estimators like the Common Correlated Effects (CCE) approach to address these dependencies (Beyene, 2022). However, future studies are encouraged to expand the variable set to capture broader structural and contextual factors.

From a policy perspective, these results suggest that G7 governments must adopt a more nuanced and evidence-based approach to economic planning. The positive influence of industrialization and trade openness should encourage policymakers to deepen trade integration, modernize industrial policies, and enhance international competitiveness. At the same time, the counterproductive effects associated with AI patenting and education expenditure highlight the need to focus not merely on the volume of investment but on its strategic deployment and institutional effectiveness.

To address the findings of this study and promote sustainable economic growth in G7 countries, several policy recommendations can be proposed. First, governments should prioritize policies that support industrialization and trade openness, such as investing in infrastructure, reducing trade barriers, and fostering international partnerships, as these measures enhance productivity and create job opportunities. Second, education systems should be reformed to focus on quality, employability, and future-readiness. This includes aligning curricula with labor market needs, investing in STEM and digital skills, improving teacher training, and addressing inefficiencies in education systems to ensure that spending translates into long-term socioeconomic returns. Third, while AI innovation holds long-term potential, its short-term disruptive effects should be mitigated through complementary policies, such as national reskilling strategies, labor market flexibility initiatives, and social safety nets, to support workers and industries during the transition. Fourth, boosting investment in infrastructure and technology is critical, and governments should encourage public-private partnerships and provide incentives for private sector participation. Special focus should be given to sectors where digital transformation and green technology overlap, to ensure that economic growth is also environmentally sustainable.

Future research should build on this study by exploring the mechanisms through which AI-related innovations and education investments influence growth, particularly across different time horizons and institutional contexts. Longitudinal and country-specific studies are necessary to assess lag effects and structural constraints. Additionally, incorporating institutional variables such as governance quality, regulatory efficiency, and labor market rigidity may offer deeper insights into the heterogeneous impacts observed among G7 nations. Expanding the analysis to include non-G7 advanced economies or emerging markets could also help generalize findings and identify broader trends.

By implementing these recommendations, policymakers can create an enabling environment for sustainable economic growth and ensure that the benefits of innovation, education, and investment are fully realized. A balanced policy mix—grounded in empirical evidence, adaptive governance, and inclusive strategies—will be critical for navigating the complex and evolving landscape of global economic development.

Ethical approval and informed consent statements

This study did not involve human participants, animals, or sensitive data requiring ethical approval or informed consent.

Data availability statement
Data can be made available upon request to the corresponding author.

Acknowledgments
The authors would like to express their gratitude to reviewers whose valuable insights helped improve the quality of this study.

References

Abdulqadir, I.M., Kibet, L. and Osoro, K. (2022), "Dynamic inference of healthcare expenditure on economic growth in Sub-Saharan Africa", *Journal of Economic and Administrative Sciences*, Vol. 38 No. 4, pp. 453-471, doi: 10.1108/JEAS-11-2020-0209.

Abid, I. (2025a), "Exploring the role of artificial intelligence (AI) innovation in economic performance: evidence from East Asia", *International Journal of Innovative Research and Scientific Studies*, Vol. 8 No. 2, pp. 3665-3677, doi: 10.53894/ijirss.v8i2.6070.

Abid, I. (2025b), "Drivers of labor force participation and economic growth in gulf cooperation countries region: a dynamic panel analysis", *International Journal of Sustainable Development and Planning*, Vol. 20 No. 5, pp. 2065-2077, doi: 10.18280/ijsdp.200523.

Abid, I. (2025c), "Balancing growth and sustainability: the role of economic factors in adjusted net savings in South Asia", *International Journal of Sustainable Development and Planning*, Vol. 20 No. 4, pp. 1487-1497, doi: 10.18280/ijsdp.200412.

Abid, I. (2025d), "Impact of technological change and economic factors on the employment level in South Africa", in Achour, Z. (Ed.), *Advanced Labor Studies Research and Modern Practice*, IGI Global Scientific Publishing, pp. 229-272, doi: 10.4018/979-8-3373-1962-9.ch008.

Abid, I. (2025e), "Economic, technological, and institutional factors affecting international tourism in MENA countries", *International Journal of Innovative Research and Scientific Studies*, Vol. 8 No. 2, pp. 3548-3560, doi: 10.53894/ijirss.v8i2.6054.

Abid, I. (2025f), "Employment trends and economic factors in MENA region: an ARDL approach", *International Journal of Innovative Research and Scientific Studies*, Vol. 8 No. 2, pp. 4334-4344, doi: 10.53894/ijirss.v8i2.6306.

Abid, I. (2025g), "The role of economic and environmental variables in green growth: evidence from Saudi Arabia", *Engineering, Technology and Applied Science Research*, Vol. 15 No. 1, pp. 20433-20439, doi: 10.48084/etasr.9836.

Abid, I. and Gafsi, N. (2025), "Economic complexity, environmental sustainability, and technological integration in Saudi Arabia: analyzing long-term trends", *International Journal of Energy Economics and Policy*, Vol. 15 No. 3, pp. 669-683, doi: 10.32479/ijeep.18806.

Abid, I., Ben Salem, S. and Frikha, W. (2024a), "Impact of political instability of monetary policy conduct and economic activity recovery: empirical investigation", *SN Business and Economics*, Vol. 4 No. 12, p. 154, doi: 10.1007/s43546-024-00750-2.

Abid, I., Hechmi, S. and Chaabouni, I. (2024b), "Impact of energy intensity and CO2 emissions on economic growth in gulf cooperation Council countries", *Sustainability*, Vol. 16 No. 23, 10266, doi: 10.3390/su162310266.

Acemoglu, D. and Restrepo, P. (2020), "Robots and jobs: evidence from US labor markets", *Journal of Political Economy*, Vol. 128 No. 6, pp. 2188-2244, doi: 10.1086/705716.

Acemoglu, D., Johnson, S. and Robinson, J.A. (2005), "Institutions as a fundamental cause of long-run growth", *Handbook of Economic Growth*, Vol. 1, pp. 385-472, doi: 10.1016/s1574-0684(05)01006-3.

Aghion, P. and Howitt, P. (1992), "A model of growth through creative destruction", *Econometrica*, Vol. 60 No. 2, pp. 323-351, doi: 10.2307/2951599.

Aghion, P. and Howitt, P. (2009), *The Economics of Growth*, MIT Press, Cambridge, MA.

Arellano, M. and Bond, S. (1991), “Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations”, *The Review of Economic Studies*, Vol. 58 No. 2, p. 277, doi: 10.2307/2297968.

Baldwin, R. and Taglioni, D. (2006), *Gravity for Dummies and Dummies for Gravity Equations*, National Bureau of Economic Research (NBER), Cambridge, MA, doi: 10.3386/w12516.

Barro, R.J. (1991), “Economic growth in a cross section of countries”, *The Quarterly Journal of Economics*, Vol. 106 No. 2, pp. 407-443, doi: 10.2307/2937943.

Barro, R. (1996), *Determinants of Economic Growth: A Cross-Country Empirical Study*, National Bureau of Economic Research (NBER), Cambridge, MA, doi: 10.3386/w5698.

Barro, R.J. and Sala-i-Martin, X. (2017), *Economic Growth*, 2nd ed., MIT Press, Cambridge, MA.

Becker, G.S. (1964), *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*, University of Chicago Press, Chicago, IL.

Beyene, A.B. (2022), “Governance quality and economic growth in Sub-Saharan Africa: the dynamic panel model”, *Journal of Economic and Administrative Sciences*, Vol. 40 No. 2, pp. 404-418, doi: 10.1108/jeas-08-2021-0156.

Blanchard, O., Olivier, J. and Giavazzi, F. (2002), “Current account deficits in the Euro area: the end of the Feldstein-Horioka puzzle?”, *Brookings Papers on Economic Activity*, Vol. 2002 No. 2, pp. 147-209, doi: 10.1353/eca.2003.0001.

Brynjolfsson, E., Rock, D. and Syverson, C. (2019), “Artificial intelligence and the modern productivity paradox: a clash of expectations and statistics”, *Economics of Artificial Intelligence*, Vol. 1 No. 1, pp. 23-57, doi: 10.3386/w24001.

Chaabouni, I. and Abid, I. (2025), “Key drivers of energy consumption in the gulf cooperation Council countries: a panel analysis”, *Engineering, Technology and Applied Science Research*, Vol. 15 No. 2, pp. 21627-21632, doi: 10.48084/etasr.10233.

Chang, H.-J. (2002), *Kicking Away the Ladder: Development Strategy in Historical Perspective*, Anthem Press, London.

Dasgupta, S. and Singh, A. (2006), “Manufacturing, services and premature deindustrialization in developing countries: a Kaldorian analysis”, WIDER Working Paper Series RP2006-49, World Institute for Development Economic Research (UNU-WIDER).

Dollar, D. and Kraay, A. (2004), “Trade, growth, and poverty”, *The Economic Journal*, Vol. 114 No. 493, pp. F22-F49, doi: 10.1111/j.0013-0133.2004.00186.x.

Easterly, W. and Levine, R. (2001), “It’s not factor accumulation: stylized facts and growth models”, *The World Bank Economic Review*, Vol. 15 No. 2, pp. 177-219, doi: 10.1093/wber/15.2.177.

Fagerberg, J. and Verspagen, B. (2002), “Technology-gaps, innovation-diffusion and transformation: an evolutionary interpretation”, *Research Policy*, Vol. 31 Nos 8-9, pp. 1291-1304, doi: 10.1016/s0048-7333(02)00064-1.

Frankel, J.A. and Romer, D. (2019), “Does trade cause growth?”, *American Economic Review*, Vol. 89 No. 3, pp. 379-399, doi: 10.1257/aer.89.3.379.

Frees, E.W. (1995), “Assessing cross-sectional correlation in panel data”, *Journal of Econometrics*, Vol. 69 No. 2, pp. 393-414.

Friedman, M. (1937), “The use of ranks to avoid the assumption of normality implicit in analysis of variance”, *Journal of the American Statistical Association*, Vol. 32, pp. 675-701, doi: 10.1080/01621459.1937.10503522.

Gonzales, J.T. (2023), “Implications of AI innovation on economic growth: a panel data study”, *Journal of Economic Structures*, Vol. 12 No. 1, 13, doi: 10.1186/s40008-023-00307-w.

Gordon, R.J. (2016), *The Rise and Fall of American Growth: The U.S. Standard of Living since the Civil War*, Princeton University Press, Princeton, NJ, doi: 10.1515/9781400873302.

Griffith, R., Redding, S. and Reenen, J.V. (2004), “Mapping the two faces of R&D: productivity growth in a panel of OECD industries”, *The Review of Economics and Statistics*, Vol. 86 No. 4, pp. 883-895, doi: 10.1162/0034653043125194.

Grossman, G.M. and Helpman, E. (1991), *Innovation and Growth in the Global Economy*, MIT Press, Cambridge, MA.

Hanushek, E.A. and Woessmann, L. (2020), "Chapter 14 - education, knowledge capital, and economic growth", in Bradley, S. and Green, C. (Eds), *The Economics of Education*, 2nd ed., Academic Press, pp. 171-182, ISBN 9780128153918, doi: 10.1016/B978-0-12-815391-8.00014-8.

IMF (2022), *World Economic Outlook: War Sets Back the Global Recovery*, International Monetary Fund, Washington, DC.

IMF (2023), *World Economic Outlook*, International Monetary Fund, Washington, DC.

Kao, C. (1999), "Spurious regression and residual-based tests for cointegration in panel data", *Journal of Econometrics*, Vol. 90 No. 1, pp. 1-44, doi: 10.1016/s0304-4076(98)00023-2.

Krueger, A.B. and Lindahl, M. (2001), "Education for growth: why and for whom?", *Journal of Economic Literature*, Vol. 39 No. 4, pp. 1101-1136, doi: 10.1257/jel.39.4.1101.

Krugman, P. (1994), "Competitiveness: a dangerous obsession", *Foreign Affairs*, Vol. 73 No. 2, pp. 28-44, doi: 10.2307/20045917.

Levine, R. and Zervos, S. (1998), "Stock markets, banks, and economic growth", *American Economic Review*, Vol. 88, pp. 537-558, doi: 10.1596/1813-9450-1690.

Lucas, R.E. (1988), "On the mechanics of economic development", *Journal of Monetary Economics*, Vol. 22 No. 1, pp. 3-42, doi: 10.1016/0304-3932(88)90168-7.

Mankiw, N.G., Romer, D. and Weil, D.N. (1992), "A contribution to the empirics of economic growth", *The Quarterly Journal of Economics*, Vol. 107 No. 2, pp. 407-437, doi: 10.2307/2118477.

Muthoni, N.M. and Osoro, K. (2023), "The dynamic effects of external debt and public investment on economic growth in Sub-Saharan Africa", *Journal of Economic and Administrative Sciences*, Vol. 39 No. 2, pp. 183-200, doi: 10.1108/JEAS-02-2022-0049.

OECD (2019), *Innovation and Growth in the G7 Economies*, Organisation for Economic Co-operation and Development, Paris, doi: 10.1007/springerreference_75881.

OECD (2021), "Education at a glance 2021", doi: 10.1787/b35a14e5-en.

Parteka, A. and Kordalska, A. (2023), "Artificial intelligence and productivity: global evidence from AI patent and bibliometric data", *Technovation*, Vol. 125, 102764, doi: 10.1016/j.technovation.2023.102764.

Pedroni, P. (1999), "Critical values for cointegration tests in heterogeneous panels with multiple regressors", *Oxford Bulletin of Economics and Statistics*, Vol. 61 No. s1, pp. 653-670, doi: 10.1111/1468-0084.0610s1653.

Pedroni, P. (2004), "Panel cointegration: asymptotic and finite sample properties of pooled time series tests with an application to the PPP hypothesis", *Econometric Theory*, Vol. 20, pp. 597-625, doi: 10.1017/S0266466604203073.

Pesaran, M.H. (2007), "A simple panel unit root test in the presence of cross-section dependence", *Journal of Applied Econometrics*, Vol. 22 No. 2, pp. 265-312, doi: 10.1002/jae.951.

Pesaran, M.H. and Yamagata, T. (2008), "Testing slope homogeneity in large panels", *Journal of Econometrics*, Vol. 142, pp. 50-93, doi: 10.1016/j.jeconom.2007.05.010.

Rodrik, D. (2004), "Industrial policy for the twenty-first century", *SSRN Electronic Journal*, CEPR Discussion Paper No. 4767, doi: 10.2139/ssrn.617544.

Rodrik, D. (2008), *One Economics, Many Recipes: Globalization, Institutions, and Economic Growth*, Princeton University Press, Princeton, NJ, doi: 10.2307/j.ctvcm4jbh.

Rodrik, D. (2015), "Premature deindustrialization", *Journal of Economic Growth*, Vol. 21 No. 1, pp. 1-33, doi: 10.1007/s10887-015-9122-3.

Rodrik, D. (2018), "Populism and the economics of globalization", *Journal of International Business Policy*, Vol. 1, pp. 12-33, doi: 10.1057/s42214-018-0001-4.

Romer, P.M. (1990), "Endogenous technological change", *Journal of Political Economy*, Vol. 98 No. 5, pp. S71-S102, doi: 10.1086/261725.

Schumpeter, J.A. (1942), *Capitalism, Socialism and Democracy*, Harper & Brothers, New York.

Solow, R.M. (1956), "A contribution to the theory of economic growth", *The Quarterly Journal of Economics*, Vol. 70 No. 1, p. 65, doi: 10.2307/1884513.

Szirmai, A. and Verspagen, B. (2015), "Manufacturing and economic growth in developing countries, 1950-2005", *Structural Change and Economic Dynamics*, Vol. 34 No. 1, pp. 46-59, doi: 10.1016/j.strueco.2015.06.002.

Westerlund, J. (2008), "Panel cointegration tests of the Fisher hypothesis", *Journal of International Money and Finance*, Vol. 27 No. 7, pp. 1196-1208, doi: 10.1002/jae.967.

Wooldridge, J.M. (2002), *Econometric Analysis of Cross Section and Panel Data*, The MIT Press, Cambridge.

World Bank (2021), *World Development Report 2021: Data for Better Lives*, World Bank, Washington, DC, doi: 10.1596/978-1-4648-1600-0.

Further reading

Ahmad, M., Jiang, P., Murshed, M., Shehzad, K., Akram, R., Cui, L. and Khan, Z. (2021), "Modelling the dynamic linkages between eco-innovation, urbanization, economic growth and ecological footprints for G7 countries: does financial globalization matter?", *Sustainable Cities and Society*, Vol. 70, 102881, doi: 10.1016/j.scs.2021.102881.

Elfaki, K.E., Handoyo, R.D. and Ibrahim, K.H. (2021), "The impact of industrialization, trade openness, financial development, and energy consumption on economic growth in Indonesia", *Economies*, Vol. 9 No. 4, p. 174, doi: 10.3390/economies9040174.

Haouas, A., Ochi, A. and Labidi, M.A. (2024), "Sources of Algeria's economic growth, 1979–2019: Augmented growth accounting framework and growth regression method", *Regional Science Policy and Practice*, Vol. 16 No. 3, 12448, doi: 10.1111/rsp3.12448.

Kanval, N., Ihsan, H., Irum, S. and Ambreen, I. (2024), "Human capital formation, foreign direct investment inflows, and economic growth: a way forward to achieve sustainable development", *Journal of Management Practices, Humanities and Social Sciences*, Vol. 8 No. 3, doi: 10.33152/jmphss-8.3.5.

LaVeist, T.A., Pérez-Stable, E.J., Richard, P., Anderson, A., Isaac, L.A., Santiago, R., Okoh, C., Breen, N., Farhat, T., Assenov, A. and Gaskin, D.J. (2023), "The economic burden of racial, ethnic, and educational health inequities in the US", *JAMA*, Vol. 329 No. 19, p. 1682, doi: 10.1001/jama.2023.5965.

Rafique, M.Z., Doğan, B., Husain, S., Huang, S. and Shahzad, U. (2021), "Role of economic complexity to induce renewable energy: contextual evidence from G7 and E7 countries", *International Journal of Green Energy*, Vol. 18 No. 7, pp. 745-754, doi: 10.1080/15435075.2021.1880912.

Raihan, A., Tanchangya, T., Rahman, J. and Ridwan, M. (2024), "The influence of agriculture, renewable energy, international trade, and economic growth on India's environmental sustainability", *Journal of Environmental and Energy Economics*, Vol. 3 No. 1, pp. 37-53, doi: 10.56946/jeee.v3i1.324.

Žalėnienė, I. and Pereira, P. (2021), "Higher education for sustainability: a global perspective", *Geography and Sustainability*, Vol. 2 No. 2, pp. 99-106, doi: 10.1016/j.geosus.2021.05.001.

Corresponding author
Ihsen Abid can be contacted at: isbaklouti@imamu.edu.sa